COMP1720/6720 *Practice* Final Exam 2019

Note

There are no answers for the "code" questions, however there is an example answer provided for the design task (Question 3).

Student ID:

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Permitted time: 15 minutes reading, 120 minutes writing

Permitted materials: one A4 page with notes on both sides

Make sure you read each question carefully.

Questions are not equally weighted, and the size of the answer box is not necessarily related to the length of the expected answer or the number of marks given for the question.

All answers must be written in the boxes provided in this booklet. You will be provided with scrap paper for working, but only the answers written in this booklet will be marked. Do not remove this booklet from the examination room. There is additional space at the end of the booklet in case the boxes provided are insufficient. If you use these extra pages, make sure you clearly label which question the answer refers to.

Greater marks will be awarded for short, specific answers than long, vague/rambling ones. Marks may be deducted for providing information that is irrelevant to a question. If a question ask for you to "explain your answer", make sure both your *answer* (e.g. yes/no) and your *explanation* are clearly indicated. If a question has several parts, you may answer the later parts even if you cannot answer the earlier ones.

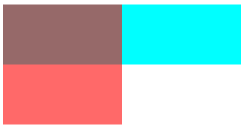
Remember the major project exhibition is on directly after this exam: **4:30pm–6:30pm** in the HN computer labs 1.23 & 1.24 (on the ground floor in building #145). There will be free pizza & (soft) drinks, so come over and see what your classmates have made (and invite your friends, too).

Question 1 p5 skills (25 marks total)

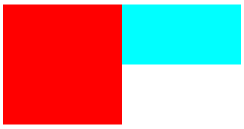
Part 1 5 marks

Assuming there's nothing else on the canvas, what will the canvas look like after the following code has been executed?

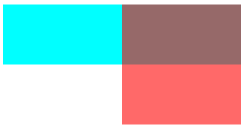
```
colorMode(RGB);
noStroke();
fill(0, 255, 255);
rect(0, 0, width, height/2);
fill(255, 0, 0, 150);
rect(0, 0, width/2, height);
```



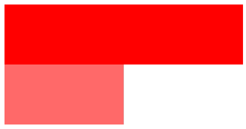
(a)



(b)



(c)



(d)

Part 2 5 marks

If there's a variable of type `Boolean`, how many **different** possible values might it have?

Part 3 5 marks

```
function setup() {
    createCanvas(800, 600);
    frameRate(10);
}

function draw() {
    ellipse(random(width), random(10*frameCount), 20, 20);
}
```

Draw what the canvas will look like after the above p5 code has been running for approximately **one** second.

Part 4 5 marks

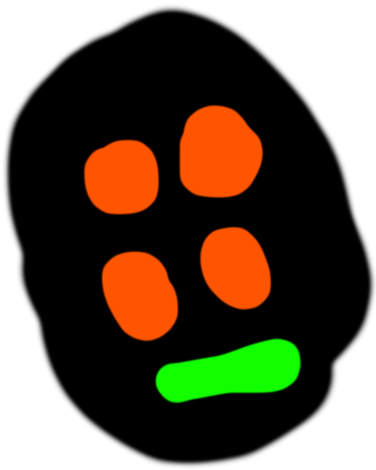
```
function setup() {
    createCanvas(800, 600);
    frameRate(10);
}

function draw() {
    ellipse(random(width), random(10*frameCount), 20, 20);
}
```

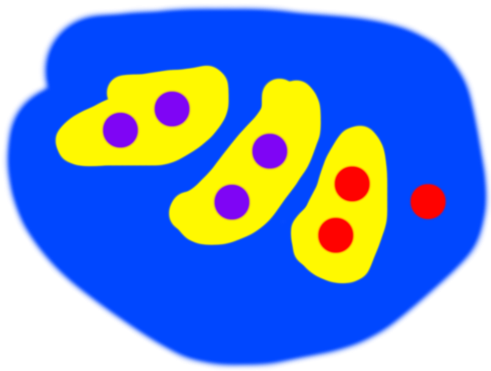
Draw what the canvas will look like after the above p5 code (the same code as in Part 3) has been running for approximately **ten** seconds.

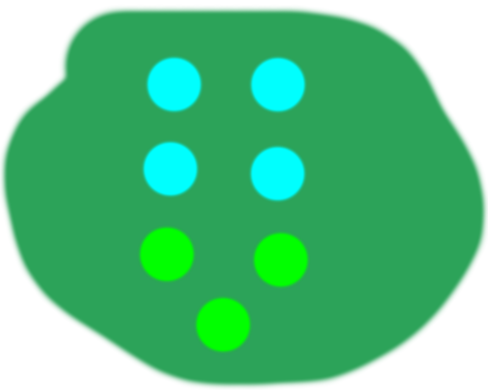
Part 5 5 marks

```
var jerp = [
  [200, 400],
  [600, 400],
  [true, false],
  true
];
```

Which of these 4 pictures **best** represents the value of the `jerp` variable?







Question 2 Film effects (25 marks total)

You've been employed the post-production team for Steven Spielblorp's new film *The Purple Jiblet* and the team is using p5 to do the film's visual effects.

This question has three parts. If you get stuck on any one part, you can still answer the next one—just continue as if you'd answered the previous part. Although you must write real code, don't worry if you forget a semi-colon here or there—you'll be marked on whether you can demonstrate that you understand the basic concepts.

Part 1 (5 marks)

Someone else on the post-production team has written a function

```
function drawMovieFrame(frameIndex) {
  // drawMovieFrame code in here
  // ...
}
```

which takes a single parameter called `frameIndex` (of type `Number`) and draws the corresponding frame of the movie to the canvas (stretched to the full width & height of the canvas). For example, to draw the first frame of the film, you would call `drawMovieFrame(0)`

Your task is to write a p5 sketch which implements a "fade-in" effect—starting with a black screen, and "fading in" to the movie as it starts playing over a period of 60 frames. After 60 frames, your code should just play the movie as-is (i.e. the fade-in effect should finish after 60 frames).

Part 2 (10 marks)

Steven Spielblorp wants an interactive p5 sketch to promote the movie (just a regular p5 sketch like you've been building all along in this course). He wants to have a bunch of purple blobs over the screen which each contain a word (written in white):

On this page & the following page a sketch "skeleton" is provided, which includes the data (positions & labels) for the blobs. **Your task** is to complete the sketch so that it uses value of the `purpleBlobs` variable to draw the picture above.

```
var purpleBlobs = [
  {
    pos: [100, 320], label: "who"
  },
  {
    pos: [150, 150], label: "is"
  },
  {
    pos: [300, 120], label: "the"
  },
  {
    pos: [550, 250], label: "Jiblet"
  },
  {
    pos: [700, 330], label: "master?"
  }
]
```

```
function setup() {
    createCanvas(800, 600);
    // your setup code goes here









}

function draw() {
    // your draw loop code goes here














}

// any additional code (e.g. functions, variables) goes here












// if you need more room, use the pages at the back of this exam booklet
```

Part 3 (10 marks)

Finally, when the viewer moves their mouse over *any* of the purple blobs, that blob must:

1. grow in size
2. fade slowly from purple to green
3. begin rotating (clockwise)

while this is happening, the other blobs must remain unchanged.

Your task is to implement this updated "dynamic" version of the sketch. You may either write everything from scratch, or refer to the code you wrote in the previous part of the question.

Question 3 Design task (50 marks)

Using one of the following themes, describe a design for an original **interactive** artwork in p5. Your sketch should communicate its main messages after about three minutes of interaction. You can use text and/or hand-drawn pictures in your answer. You don't *have* to draw pictures/diagrams, but feel free to use them wherever they help you communicate the design of your sketch.

As part of your description, address the following questions:

- what are the main ideas and/or emotions that you are trying to communicate?
- how will interaction enhance the artistic effect of the design?
- how does your design use affordances and feedback?
- where is the *duende*? i.e. how will yours stand out from all the other works based on the same theme?
- how would you create your design using javascript and p5?

1720 themes

> Note
>
> For **COMP1720**, these are the actual themes which will be on the 2019 final exam.

If you're an *undergraduate* student in COMP1720, choose **one** theme from the following list:

1. take a chance on me
2. gimme gimme gimme
3. dancing queen

6720 themes

> Note
>
> For **COMP6720**, you won't know the themes until you get into the final exam.

If you're a *masters* student in COMP6720, choose **one** theme from the following list:

1. and the winner is…
2. Mother is always right
3. into the beard

Note: you don't have to use all of these pages for your answer—the extra pages are included in case you need them for other questions (as described on the title page).

Note: this is an *example* answer

It's not meant to be a the best possible answer—I'm sure you can do better than this. It's here to show you what an answer might look like for a "Swooping Magpie" theme. This theme won't be on the exam, however—for the themes on the *actual* final exam see page 11.

As always your answer to this question in the exam must **be original and be your own work**. So be careful about studying as a group. If there are a number of answers which are very similar to each other the examiners have the right to interview the students concerned, and the usual academic integrity policies & penalties apply.

Theme: Swooping Magpie

I choose the "Swooping Magpie" theme. It is a real **shock** to be swooped by a magpie. People are often swooped when riding their bicycles. But the magpies are not evil—it is just a part of nature that they feel as though they are protecting their babies. In my design, I will communicate:

- the shock of being swooped
- that magpies are a part of nature

My setting will be someone riding their bicycle. I will have an audio background of magpie calls and the creaking of a bicycle chain. I will also have some **humour** in my sketch and there will be a message about nature.

Opening scene

My opening scene will be a simple scene with someone riding a bicycle up a gentle hill. The person will be riding slowly. If you move the mouse they will ride slightly more quickly. (Actually, it is possible that moving the mouse one way will make them go forwards and the other way go backwards; I would try these ideas out when building the sketch.) You will hear magpie calls and the bicycle.

Shock! Magpie Swoop!

Suddenly, the scene will flash to black as a bird shape covers the canvas and then shrinks to be an easily-seen magpie. The magpie will zoom down and hit the rider on the head and they will fall off. The scene will then close with panels sliding forward from the left and right with the new scene drawn on them. This is like the curtains on a stage closing an act. There will be a loud squawk and a "thunk" on the audio track as the magpie hits its victim.

Accessories for rider and magpie

The next scene will be like a game where you can drag and drop items to provide accessories to the rider and the magpie. There will be large pictures of the rider (on the right panel) and the magpie (on the left panel) and a horizontal tool-bar across the bottom of the screen. The items that you can drag and drop on to the rider or the magpie are

- a big stick
- a sign saying "beware magpie"
- sun glasses
- some worms

You can only give the stick to the rider. You can only give the worms to the magpie. You can give the sign and the sun-glasses to either the rider or the magpie. You only have one choice. If you make an illegal choice, then the accessory jumps back to the tool-bar and an error message is printed out. When you have dragged the accessory to its target, then this scene closes (with closing panels as before) and the sketch returns to the riding scene again.

Note that either the rider or the magpie can wear sun-glasses! A bird wearing sun-glasses is meant to be funny. The audio track in this scene would be a "la di da" game show theme, perhaps overlaid with some magpie chirping.

Back to the swooping scene

Now we are back in the country-side watching our victim ride his/her bicycle up the hill. Except now there will be a new set of outcomes. After each outcome, this scene will close and the sketch will return to the "accessories" scene. The possible outcomes are:

- If the bicycle rider has the stick, then the magpie swoops, but does not hit the rider. But the rider still wobbles and falls off the bicycle because it is hard to ride a bicycle while holding a stick!
- If the rider or the magpie are wearing sun-glasses, then the magpie still swoops and makes the rider fall off.
- If either the rider or the magpie were given the sign, then the sign would appear in the sketch. The rider will stop to read the sign and then the magpie will swoop down and hit the rider while they are reading it.

- If the magpie was given the worms, then it will swoop down but land next to the rider and try to be the rider's friend. But then another magpie will swoop down and hit that magpie and knock it over. (The message here is that we should leave nature alone.)

In this sketch, interaction makes it playful

In addition, the user gets to imagine what the outcomes of the different accessories might be. Interaction **slows down** the message of this sketch. In fact, it is likely that the message with the worms will be something of a punch-line for this sketch and, if so, it could have considerable impact. Of course, if the user does not choose the worms last it will not be as much of a punch-line.

This sketch is feasible in p5.js

The biggest challenge would be constructing a good animation of the bicycle riding and the rider falling off. I would use cartoon images for the rider and a simple schematic for the bicycle; but this will be challenging. The magpie would be an image with various amounts of dynamic scaling to shrink it down and hit the rider. The audio track would be easy to construct from field recordings or to come by from the internet.